



A multimodal emotion recognition system using deep convolution neural networks

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ABSTRACT

Despite the progress in computer-technology, with regard to Human-Computer Interaction (HCI), emotion recognition is still a challenging problem. In this paper, we present a novel multimodal emotion recognition system capable of recognizing emotions from audio, video, and text data using deep convolution neural networks. The system is able to recognize happy, angry, sad, afraid, disgust, surprise and neutral emotions. We used three datasets to train and test the system, one set for each of the three input formats. The results show a recognition accuracy rate of 100% for audio, 69% for video, and 64% for text. When applying the decision-level fusion, the recorded accuracy rate is 80%. These results confirm that the system is effective in recognizing human emotions.

Introduction

Emotion is a way for humans to express feelings, wishes, needs, interests, and concerns. They are being used as a communication tool between different living species. As computers continue to evolve in the AI era, they have become an important part of our lifestyle. Yet, they need to recognize human emotions to provide better responses to their interactions. For example, welfare services can be improved by recognizing the elderly people's condition. Also, emotion recognition may facilitate the automatic diagnosis of mental diseases. Emotion-aware chatbots have started to help us execute our daily activities smoothly [41]. Healing conversations is another research area that can benefit from speech emotion recognition. These days software developers are encouraged to create applications that adjust the type of goods and services they offer according to the detected emotional state of the customers or even create intelligent machines that have motions on their own, like the school teaching robots or the healthcare providing robots. Therefore, researchers have become interested in the recognition and classification of human emotions that emerge from different sources such as facial expressions, speech, and body language [1]. Usually, human emotions can be recognized by various modalities such as unimodal behavior (e.g. facial expressions, speech, eye gaze, hand gestures, or body postures) ([5]; [25]; [34,38,39,42]) bimodal behavior (e.g. both facial expressions and speech or speech and text) [20,25,27,31, 47], or by multimodal behavior (e.g. facial expressions, speech, and text all together) [16,32,35,44]. The emotion recognition modalities are

illustrated in an AND/OR tree shown in Fig. 1. In this tree, one-connectors emerge from OR nodes, whereas k-connectors emerge from AND nodes [33].

The conventional approach for handling emotion recognition depends on unimodal behavior, like analyzing either video or audio segments. Emotion recognition systems that use this approach exhibit impressive performance in specific situations. Relying solely on unimodal behavior may not always classify the emotion correctly. The unimodal approach is vulnerable to noise and data input variability. For example, camera angles and lighting conditions can affect the interpretation of facial expressions. Similarly, physiological signals can be affected by variables like physical activity and stress, impacting emotion recognition accuracy [43]. Thus, researchers' attention has been shifted toward the multimodal behavior approach. Mehta [30] wrote that different modalities have different features associated with them. To correctly classify complex human emotions, it is vital to consider their inter-relationships.

Multimodal Emotion Recognition (MER) refers to the recognition and classification of human emotions by combining different signals coming from text, speech, and face cues. It involves research related to medicine, security, AI, and human-computer interaction (HCI) [40]. Emotions can be extracted from various sources such as physical information (e.g. video, audio, and text) and physiological signals (e.g. EEG and ECG). The feature that distinguishes unimodal behavior from bimodal or multimodal behavior is the fusion of modalities. Fusion means generating a combined representation that outlines how different

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modalities interact and/or collaborate to reach the desired objective. Two common fusion strategies used in emotion recognition are feature-level fusion (aka early fusion) and decision-level fusion (aka late fusion). The emotion predictions from the different modalities can be combined using several integration approaches such as averaging, majority voting, highest accuracy, signal variance, weights based on channel noise, or optimization modeling [40]. For instance, at the decision-level fusion researchers classify emotions using weight-based criterion to reduce - if not eliminate - the discrepancies between the predicted outcomes of the modalities.

The existing fusion approaches improve multimodal information processing, yet they are highly customized. In some cases, the modalities do not agree on the same predicted emotion for the same input scenario. This is due to many reasons such as missing modality data, inter-class variations, or intra-class similarities, etc. Hence, fusion strategies are tailored to compute the best-predicted outcome. Other novel but not as popular fusion strategies include data-level fusion, the multimodal interactive attention network (MIA-Net) fusion [31], and hybrids of different fusion strategies such as the combination of feature-level fusion and decision-level fusion. Recent work showed that the decision fusion approaches for incorporating multimodal predictions into a combined prediction have outperformed predictions obtained by unimodal approaches [14]. In this research, the major contributions include (-) achieving high accuracy in emotion recognition by leveraging video, audio, and text features with the help of deep convolution neural networks, (-) highlighting the relationships between the features of different modalities, (-) using decision-level fusion to ensure accurate system performance.

When confronted with large datasets, deep learning surpasses traditional machine learning methods. Unlike conventional approaches that analyze data linearly, deep convolutional neural networks employ hierarchical functions, enabling nonlinear data processing. For intricate tasks such as facial recognition, as well as video and audio identification, deep convolutional neural networks are recommended. Given the simultaneous handling of three distinct input data formats, integrating deep convolutional neural networks into our methodology for human emotion recognition appears evident. This study utilizes seven fundamental human emotions to predict the scenarios within the input data files: disgust, anger, happiness, sadness, fear, surprise, and neutrality.

The remainder of the paper is organized as follows. The following section discusses the related works. Section 3 Covers the methodology followed, which consists of the description of the datasets used, the features extraction, and the classification processes applied in each modality. Section 4 explores the experimental results. The last section

concludes the work and highlights the future directions.

Related works

With the big success of deep learning – particularly the success of convolution neural networks (CNN) – in emotion recognition and classification, several attempts for developing deep learning-based models with promising performance are found in the literature. Some follow the unimodal approach while others follow the bimodal or the multimodal approaches. To name some of these attempts, Ma et al. [27] proposed a cross-modal noise modeling and fusion methodology over bimodal audio and images using 2D and 3D convolutional neural networks for detecting emotions from facial expressions in an image sequence. Deep convolutional neural network (DCNN) was also utilized by [49] for speech emotion recognition using discriminant temporal pyramid matching (DTPM) in speech emotion recognition tasks [49].

Navarathna et al. [32] attempted to recognize audio-visual speech in vehicles using the AVICAR database. Their visual-audio fusion results reported approximately 70% accuracy. Bhaskar et al. [6] introduced a hybrid approach for emotion classification of audio conversations based on text and speech mining. their system gave an average accuracy of 90%. Adil [2] proposed facial emotion detection using a CNN, he used the network to extract a histogram of oriented gradient features and scale-invariant feature transform from images found in the MELD dataset. To improve the accuracy and avoid data overfitting, he used batch normalization and dropouts. Thus, the suggested CNN model gave 80% accurate results for 5 emotions namely happy, sad, angry, fear, and neutral. Liliana used a deep CNN to obtain better performance when detecting emotions. She worked with 8 facial classes to train her model using different data sizes. Thus, her model gave an average accuracy rate of 92% [22]. In addition, Abdusallam et al. [1] developed a system for emotion recognition based on facial expressions and speech using deep CNN, their system gave an average accuracy of 93%. In Rafeh et al. [37] suggested a hybrid approach for emotion classification from video and audio data, their system gave an average accuracy of 99%.

The number of features processed by multimodal approaches exceeds the number of features processed by unimodal, which may improve the effectiveness of emotion recognition [23]. This is because multimodal emotion recognition technology fully considers the interaction between text, speech, image, and other modalities. At present, the modalities used in MER mainly include facial expression, speech, text, and physiological signals. Many of the attempts are hybrid combinations of different modalities to achieve the emotion recognition objective [25]. In her proposed adaptive fusion method that aggregates the results of

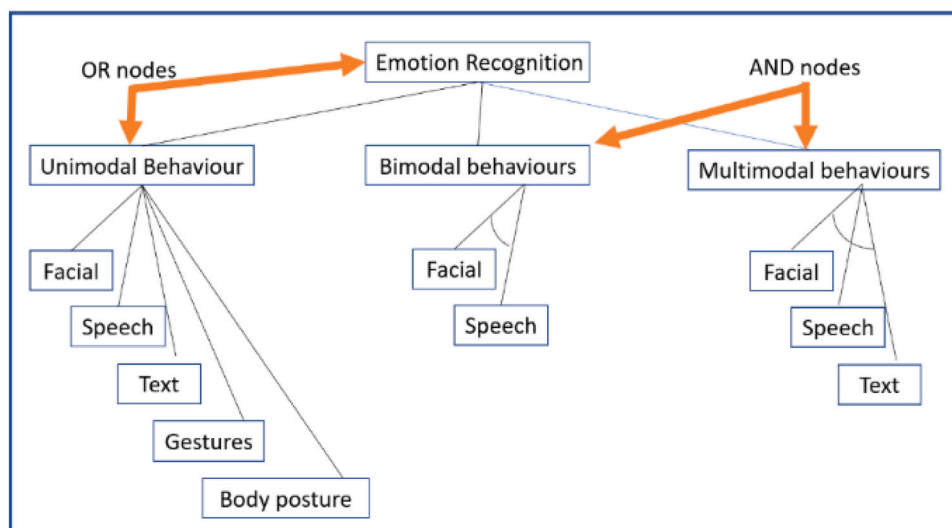


Fig. 1. AND/OR tree showing types of emotion recognition modalities.

predictions obtained for a set of classifiers, [36] computed a joint prediction using a group of classifiers with the help of a multiplicative weighting method where weights are assigned iteratively. Wang et al. proposed a hybrid fusion model that considers correlations between the features extracted from different modalities. They combined feature-level fusion and decision-level fusion, and the final emotion prediction is computed based on a weighting scheme [46]. A comprehensive review of emotion identification systems with underlining basic neural network classification models is described by Gravina et al. [13]. They considered a methodical classification of the literature on data-level, feature-level, and decision-level multimodal fusion approaches.

Methodology

This research analyzes and improves the robustness of emotion recognition, and proposes a novel approach that optimizes traditional deep neural networks that fall into poor local optima when optimizing the weightings of the deep neural network using standard methods. Moreover, the main objective of this research is to make the machines that we use in our everyday life more intelligent and user-friendly by giving them the ability to understand our emotional states and respond appropriately. For example, when the human is felling down, to cheer him/her up, the machine can tell a joke, or when the human is angry, you don't want the machine to give the wrong responses. Thus, we want the machine to act in real-time, this is the unique feature of our classification methodology. The light weight of the fusion algorithm makes the machine response almost momentarily. Yet, this classification methodology and in fact all other emotion recognition methodologies are input dependent. In other words, they rely on the input format whether it is an image, voice or text. Imagine a person singing happy birthday with an angry face. What is the expected outcome of any emotion recognition tool, will it be happy or angry?

Our proposed system comprises three simultaneous modules written in Python. One for recognizing the face emotion, one for recognizing the voice emotion, and the last for recognizing the emotion related to the uttered sentences. It is a widely accepted notion that the more input data, such as audio, video, speech, and posture, you process, the better recognition you can achieve. However, adding more variables can complicate the classifier's job, which presents a significant challenge, particularly when databases are limited in size. In the modern world, emotion recognition of humans is procuring huge interest in many areas such as bio-metric security, HCI, etc. Thus, our system can be used in many applications such as education, airport security, medical diagnosis, patient care, video gaming, employee safety, car safety, and fraud detection. Our system can be generalized to be utilized in all of these authentication related applications as long as the application has access to the Internet and can interface with the Python.

The testing datasets

MER requires synchronized modalities where features are extracted from data streams capturing events performed by the users with the help of deep CNN. Several popular MER datasets have been developed to facilitate research and experimental progress in the field, each of which has its unique advantages and limitations that require further attention. An in-depth review of these datasets hosting facial expressions, speech, and text information can be found in [21]. In this work, we used three datasets for training and testing the proposed system, one for images, one for speech, and one for text. The objective of our work is to develop a MER system that can recognize and classify emotions using deep convolution neural networks. The training data must come from video and audio files of different people, having a dataset including several data files for each emotion and subject. We divide the data files of each dataset into two sets: training and testing. The training set contains 80% of the data files and the testing set contains 20% of the files, but these

numbers are subject to change according to the quality, quantity, and contextual significance of the data which may affect the performance of deep learning models.

One of the obstacles in developing deep learning models is the lack of clean data. To overcome this obstacle, we trained the system on three specialized datasets namely the Facial Expression Recognition (FER-2013) for images [11], the Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) for audio [24], and the Emotion TEXT Dataset (ETD) for text [19]. The first dataset has two folders: a training folder containing 28,709 images and a testing folder containing 7178 images. FER-2013 is one of the most challenging facial expression recognition datasets available online, due to its inter-class variations and the imbalanced nature of its different emotion classes. The number of sample files for each emotion class is shown in Fig. 2. The second dataset contains 7356 speech and song files of 24 professional actors (12 female, 12 male), vocalizing two statements uttered in a North American accent. The speech files in the dataset explore calm, happy, sad, angry, fear, surprise, and disgust emotions, and the song files contain calm, happy, sad, angry, and fear expressions. The first class (i.e. calm) is dropped to make the dataset's classes consistent with those of the other two datasets. Each expression is produced at two levels of emotional intensity (normal, strong), with an additional neutral expression. The third dataset has an Excel file consisting of 34,793 text sentences. The data is further divided into the following 7 classes of emotions: joy (11,045 samples), sad (6722 samples), fear (5410 samples), angry (4297 samples), surprise (4062 samples), neutral (2254 samples), and disgust (856 samples). The number of sample files for each class is shown in Fig. 3. Once the data is loaded into memory, we perform preprocessing steps to clean the data; these steps include the removal of user handles, hashtags, non-ascii characters, special characters, URL addresses, and all emojis if any. The data is then split into 80% for training and 20% for testing.

Addressing imbalanced datasets is pivotal in machine learning to mitigate biased or misleading classification outcomes. Various strategies exist to rectify this issue, including employing resampling techniques such as oversampling (e.g., augmenting the minority class with synthetic or duplicated instances), under-sampling (e.g., reducing the majority class by randomly eliminating samples), utilizing the SMOTE (Synthetic Minority Over-sampling Technique) method (e.g., augmenting the minority class by interpolating between existing samples), or implementing data augmentation (e.g., introducing variations to existing samples to bolster diversity, particularly in the minority class). Alternatively, one can opt for classification algorithms inherently resilient to class imbalance, such as Support Vector Machines (SVM) with appropriate class weights or decision trees, or leverage ensemble algorithms like Random Forests or Gradient Boosting, which inherently handle class imbalances better than individual models. In evaluating our system, we employed several classification algorithms, including Support Vector Machine (SVM), Random Forest (RF), and Stochastic Gradient Descent (SGD), all of which demonstrate inherent robustness to imbalanced datasets.

Features extraction

A) Video Features: Paul Ekman asserted that facial expressions are universal in various cultures including preliterate cultures [12]. He confirmed the existence of similarities in how people express emotions on their faces. The extraction of facial expressions from the window of the real world is necessary for facial emotion recognition (FER). Yu and Zhang [48] proposed image-based static facial expression recognition with multiple deep network learning. Another approach was presented by Zhang et al. [50] in which they constructed a fusion network built with a deep belief network (DBN) to take full advantage of a 2-stream CNN, demonstrating its effectiveness in capturing spatiotemporal information. In this work, a deep CNN is proposed to detect emotions from the extracted features. We used a Python library (TensorFlow) to identify 64 landmarks on any given face. We train the deep CNN model with gray-scale images to speed up the computations. Finally, batch

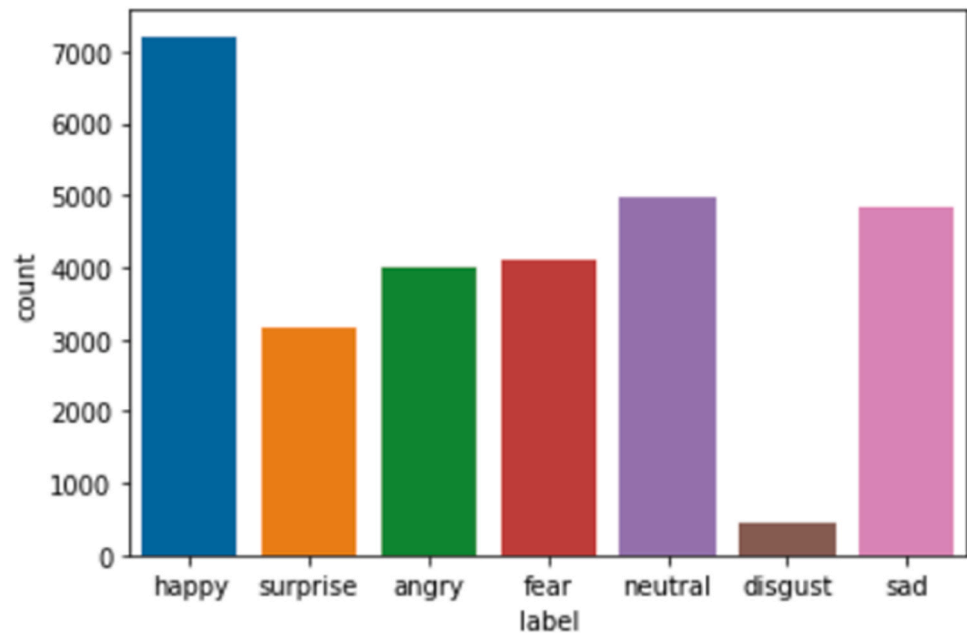


Fig. 2. The sizes of emotion classes in the FER-2013 dataset.

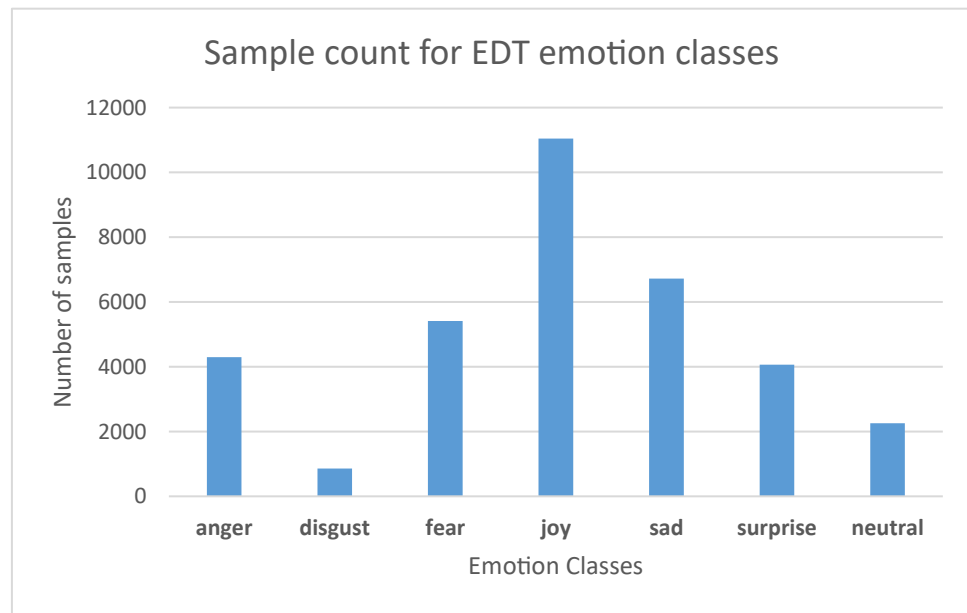


Fig. 3. The sizes of emotion classes in the ETD dataset.

normalization and dropout are applied to reduce the chance of model overfitting and improve the accuracy rate.

B) Audio Features: When classifying emotions through voice signals, researchers often consider acoustic sound features such as pitch, energy, and the speed of speech. Findings show that emotional changes highly affect individual voice features such as the basic frequency (i.e. sound pitch) [28]. In this study, seven emotion classes were studied for recognizing emotions namely: happy, sad, angry, fear, disgust, surprise, and neutral. Different variations are created in human voice features such as pitch, energy, and speed of speech in various emotional states. The emotion classification involves separating the data into a training set (80% of the data) and a testing set (20% of the data) respectively. We used one of the standard speech extraction methods, namely the Mel-Frequency Cepstral Coefficient (MFCC) to extract audio features

from speech utterances. These features include energy, spectral centroid, pitch, non-silence ratio, and volume standard deviation to generate an audio feature vector [4]. In this work, we recorded the value of energy per frame of voice, the shape of vocal cords and how they vibrate, and the speed by which the wave signal traveled. The system architecture to recognize emotions from the audio stream is shown in Fig. 4. It shows the raw audio data stream, removes non-speech noise (i.e. removes unnecessary gaps from audio), and performs a 3-sec window-based segmentation using an audio buffer.

The DCNN consists of an input layer, hidden layers and a SoftMax output layer to capture the associations between the features from the sound modality and classify them into emotion classes. These hidden layers are a combination of Conv2D layers and dense layers activated by Adam at a learning rate of 0.001 and were provided to the SoftMax

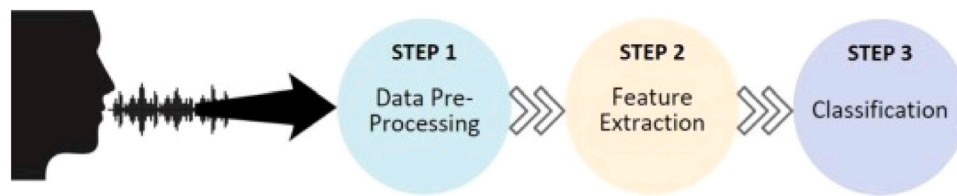


Fig. 4. Architecture of Emotion Recognition from Audio Streams.

classifier layer. The output of the SoftMax layer represents a vector of size equivalent to the seven emotions with their corresponding probabilities.

C) Text Features: For the text analysis, we used the nltk and neartext python libraries to extract text files from the audio files and performed the following preprocessing actions: Tokenizing the text file, cleaning and standardizing the data using regular expressions, deleting user handles, stop-words, hashtags, punctuation, non-ascii characters, special characters, addresses of websites (URLs) and emojis, and finally lowercasing the tokens. After cleaning the data, we are left with 34,792 lines of text. Then, we split the data into training data set (27,833 lines of text) and testing data set (6959 lines of text). Next, we developed several models using the following classifiers: Logistic Regression (LR), Support Vector Machine (SVM), Random Forest (RF), Stochastic Gradient Descent (SGD), Multi-Layer Perceptron (MLP), Multinomial Naïve Bayes (MNB), Decision Tree (DT). In the end, we adopted the classifier with the highest accuracy score.

Experimental analysis and comparison

In this subsection we present the performance of the three proposed models on the above datasets. To ensure unified fair investigation and to avoid unjust perception confusion, we compare the performance of each model with the performance of other related methods that used the same dataset. For the FER-2013 dataset, Boughrara et al. [8] introduced a model employing a multi-layer perceptron featuring a single hidden layer. They adopted an incremental training approach to ensure comprehensive learning of all data from the training dataset, achieving an 84.58% accuracy rate on the FER2013 dataset. Christou & Kanojiya [9] investigated a CNN model, achieving an accuracy of 91.12%. Ionescu et al. [17] proposed a computer vision technique for classifying human facial expressions using low-resolution images. They modified a bag of visual words model, achieving a 67.48% accuracy rate. Minaee et al. [31] introduced a deep learning method utilizing a CNN, focusing on critical facial regions and achieving accuracies of 70.02%. Verma & Verma [45] developed a hybrid deep learning CNN model for emotion recognition from images containing human faces, targeting happy, sad, surprised, and angry expressions. They applied a dropout rate of 0.2, resulting in a maximum accuracy of 97.07%. In contrast, our model achieved a 69% accuracy rate.

For the RAVDESS dataset, Luna-Jiménez et al. [26] attained an 80.08% accuracy on the RAVDESS dataset through a subject-wise 5-CV evaluation, categorizing eight emotions. Bilotti et al. [7] delved into emotion recognition via a multimodal strategy leveraging convolutional neural networks (CNNs) with multiple inputs. The highest accuracy obtained on the RAVDESS dataset was approximately 95.5%, while our model achieved a perfect 100% accuracy rate. In the case of the EDT dataset, [18] achieved an accuracy rate of 64.7%, while our model achieved a slightly higher accuracy rate of 64.8%. In summary, our models demonstrated comparable, if not superior, performance when compared to recent studies.

Decision-level fusion and emotion classification

The emotion classification in our system is performed by analyzing the video, audio, and text features separately, and combining the

predicted outcomes (i.e. decision labels) into a final proposed emotion prediction. This fusion process can be performed at the data-level, features-level, or the decision-level. In this work, we used the decision-level approach to combine the predictions. In this approach, the decision labels and their accuracy scores, obtained from the multiple modules are combined in a way that deals with disambiguation among the different modalities to obtain the final emotion decision label. By accuracy, we mean the percentage of correctly predicted quantity in the total quantity. Accuracy is the most used index for performance evaluation. Most of the MER repositories are constituted for video and audio modalities. However, very few of them have considered additional modalities for emotion recognition from text and body language [49]. We trained one model per dataset in our experiments. The accuracies from the three modalities can be fused into a single score using one of three methods: average, weighted, and majority voting. The average approach simply computes the average of the video, audio, and text accuracies. The weighted approach assigns weight to each modality that reflects the importance of that modality in the communication equation. Of course, the sum of the weights must equal to one. Mehrabian and Ferris [29] suggested that communication between human beings relies on facial expressions 55%, speech 38%, and 7% of any other modality. In our work, we adopted the Mehrabian weights according to the following equation:

$$\text{Final Accuracy} = w1 * Af1 + w2 * Af2 + w3 * Af3$$

where Af1 = Video Accuracy score, Af2 = Audio Accuracy score, Af3 = Text Accuracy score, w1 = 0.55, w2 = 0.38, and w3 = 0.07. The weights represent the confidence of the classifiers during the classification process in the combination method. Fig. 5 shows the architecture of our multimodal emotion recognition system.

Experimental results

We adopted the TensorFlow framework for our model building and training. The framework parameters were almost the same for all the models. We also used the accuracy evaluation index to evaluate the performance of the system. However, we kept the architecture and the learning parameters the same in all the experiments. Each model was trained for 100 epochs with a learning rate of 0.001. The loss function in this work is simply the summation of two values: the classification loss (cross-entropy) and the regularization weight λ (tuned on the validation set). It took around one hour to train each model using the GPU capabilities of the installed graphics card. We ran the experiments using Windows 11 running on a 13th Generation Intel(R) Core(TM) i7-1355 U Processor, 1700 MHz, 10 Core(s), 12 Logical Processor(s), 16 GB RAM, and NVIDIA GeForce MX-550 graphics card.

Face Expression Recognition: The data size is 20000 sample text sentences divided into training data, (train_size=80%) and testing data (test_size=20%). The DCNN has three layers in the sequential model. For the loss, we used sparse categorical cross-entropy and Adam optimizer. We tried other optimizers like the RMSprop optimizer proposed by Geoffrey Hinton (the father of back-propagation) and Stochastic gradient descent (SGD), but Adam gave the best performance. Also, we tried different learning rates: 0.001, 0.005, and 0.01, these rates had some effect on the training accuracy and loss but little effect on the

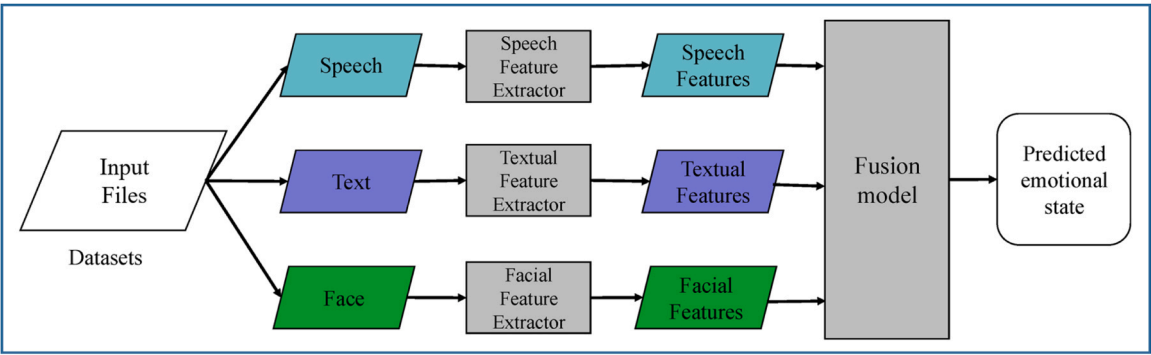


Fig. 5. The architecture of our multimodal emotion recognition system [21].

validation accuracy and loss. Moreover, we studied the effect of the number of recognized emotions on the performance of the model. The validation accuracies were 93%, 85%, and 69% when the model analyzed 2 emotions, 4 emotions, and 7 emotions respectively. Fig. 6 shows the system performance during the training phase. The accuracy for the training phase was 70% and the loss was 82%. The accuracy for the testing phase was 69% and the loss was 88%. The confusion matrix of the model (shown in Fig. 7) revealed that the model makes more mistakes for the emotion classes with fewer samples such as fear and disgust.

Speech Emotion Recognition: The audio data size is 2880 wave files, and they are divided into training data, (train_size = 80%) and testing data (test_size = 20%). After the preprocessing step that corresponds to cleaning the data by removing the noise and the unrecognizable recordings, the remaining data size is 1440 files. The DCNN has three types of layers (input, hidden, and output) in the sequential model. For the loss, we used categorical cross-entropy and Adam optimizer. Fig. 8 shows the model performance during the training and testing phases for the audio data. The accuracy for the testing phase was 100% and the loss was zero percent.

Text Emotion Recognition: The data size is 34,792 sample text sentences divided into training data (train_size = 80%) and testing data (test_size = 20%). The system attempted 7 classification models and adopted the one with the highest accuracy score. The accuracies of these classifiers are illustrated in Fig. 9. It can be seen that the logistic regression classifier has the best accuracy performance with 64%. To make sure that we selected the classifier with the highest accuracy, we changed the dataset to another dataset used for Tweet Emotions Analysis. It is called “text_emotion.csv” and it contains 40,000 tweets grouped into 12 emotion classes. Again, the dataset was divided into

training data (train_size = 80%) and testing data (test_size = 20%). We repeated the experiment in the new dataset and we obtained the following results: Logistic Regression (LR) 36%, Support Vector Machine (SVM) 36%, Random Forest (RF) 32%, Stochastic Gradient Descent (SGD) 35%, multi-layer perceptron (MLP) 30%, Multinomial Naïve Bayes (MNB) 32%, and Decision Tree (DT) 27%. It can be seen from the two experiments that the LR classifier returned the highest accuracy.

Conclusion and future work

This paper introduces a novel multimodal emotion recognition system leveraging deep convolutional neural networks. Our proposed system seamlessly integrates inputs from video, audio, and text data files, enabling the prediction of emotions for each respective source. Evaluation results showcase impressive recognition accuracy rates: 100% for audio, 69% for video, and 64% for text. Notably, upon implementing decision-level fusion, we achieve an overall accuracy rate of 80%, a commendable achievement considering the data imbalance issue and the complexity of analyzing seven distinct emotion classes compared to systems focusing on subsets of these classes.

Our system promises to yield valuable statistical insights that underpin its emotion prediction capabilities, thus facilitating expedited decision-making processes. Looking ahead, our aspirations extend to broadening the spectrum of recognized emotion classes and conducting multifaceted analyses across various emotional states. This comprehensive approach encompasses text, audio, visual imagery, and other emotion-conveying mediums, with the potential to uncover latent insights and contribute to the development of a nuanced ontology for enhanced emotion recognition. The implementation of our multimodal deep learning approach for emotion classification is not just important

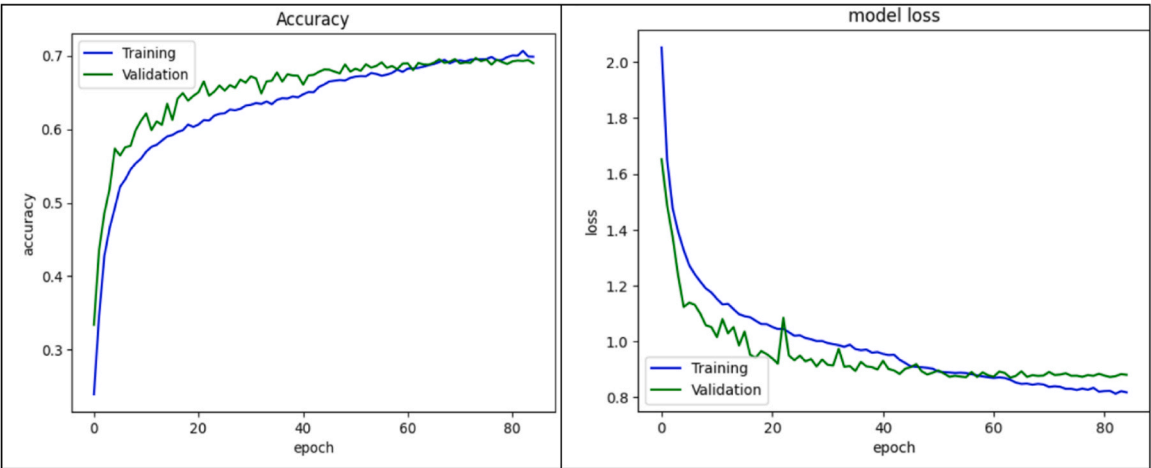


Fig. 6. Image-based model performance using the Adam optimizer and 0.001 learning rate.

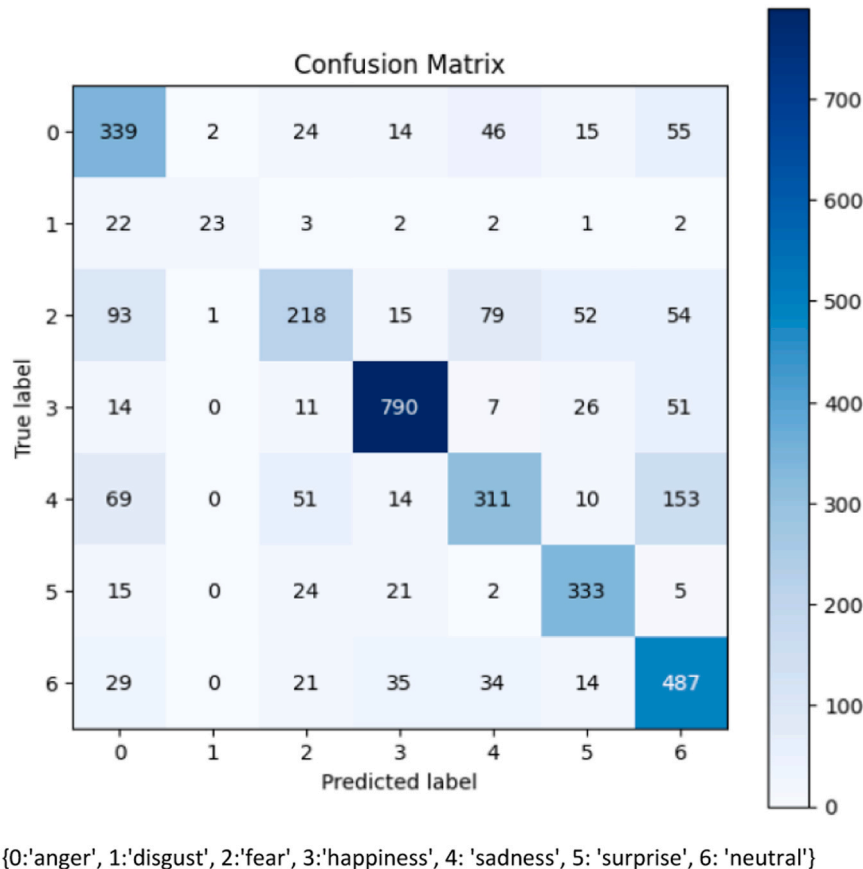


Fig. 7. Video modality confusion matrix for the 7 emotion classes.

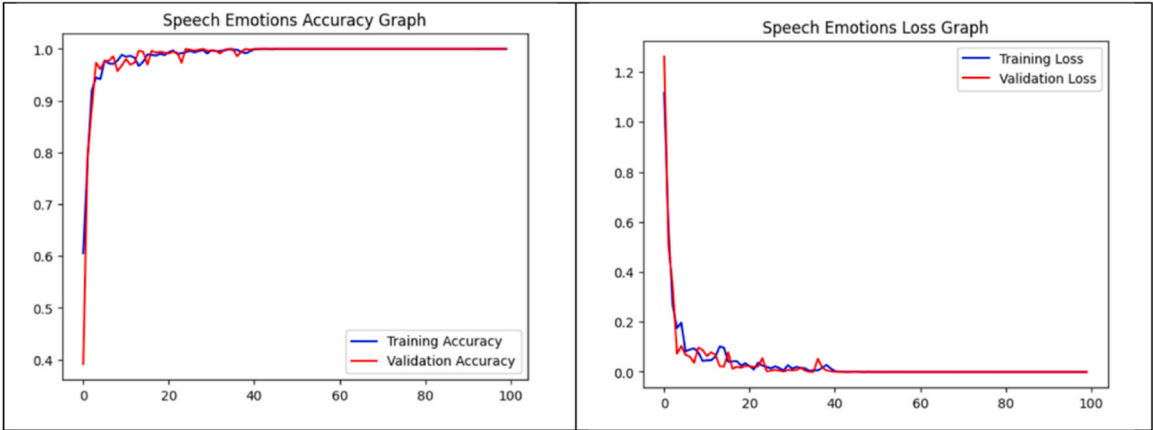


Fig. 8. Audio modality performance using the Adam optimizer and learning rate of 0.001.

but also urgently necessary for immediate deployment in real-world scenarios. Looking ahead, our plan is to integrate this approach with applications designed to handle large crowds in real-time. These applications encompass a wide range of functionalities, including but not limited to a sophisticated face-recognition system aimed at locating lost individuals within densely populated areas [15], analyzing asymmetric interactions among agents within crowds [10], and predicting cyber-attacks through the real-time detection of relevant patterns on Twitter [3]. Furthermore, our system is poised to generate valuable statistical insights to support its recognized emotion prediction, facilitating rapid decision-making processes.

Addressing linguistic-specific challenges, such as recognizing

emotions across diverse languages and accents, represents a key focus area for future research endeavors. Additionally, we intend to integrate our system with applications tailored for real-time deployment in large crowd settings. The resulting statistical information generated by our system will be instrumental in facilitating swift decision-making processes.

CRediT authorship contribution statement

Mohammed A. Almula: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing, Validation.

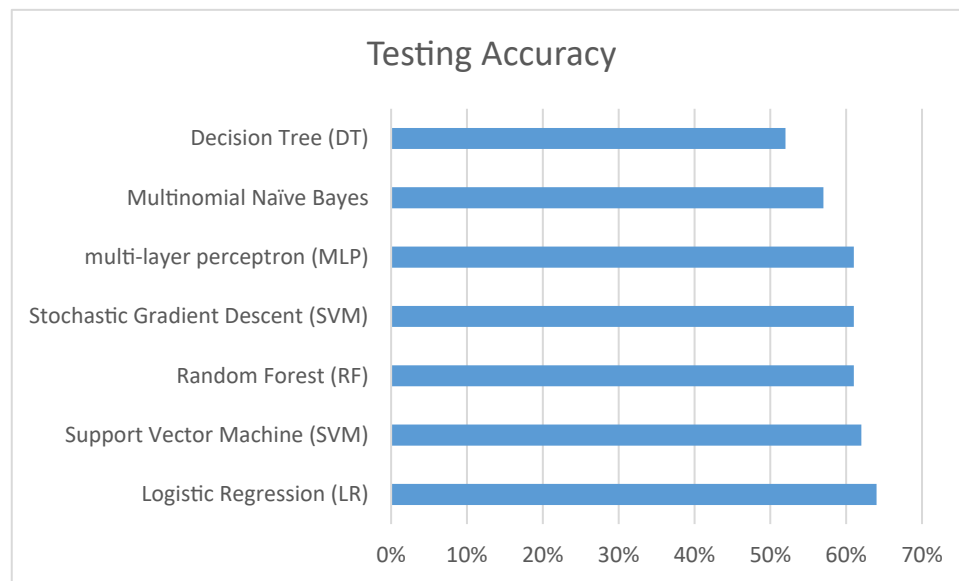


Fig. 9. Accuracy of the Text Classification Models.

Declaration of Competing Interest

I (Mohammed A. Almulla) hereby declare that the disclosed information in this article is correct and that no other situation of real, potential, or apparent conflict of interest is known to me. Also, I undertake to inform you of any change in these circumstances, including if an issue arises during the preparation of the work itself.

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